

Nikhil Tammalla

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SUMMARY

Final-year Computer Science student with hands-on experience in data science, machine learning, and applied AI. Published in IEEE and ICSSCC across 3 projects. Proficient in Python, SQL, and LLM pipelines.

EDUCATION

Amrita Vishwa Vidyapeetham — GPA: 7/10	Bangalore, KA
<i>B.Tech in Computer Science Engineering</i>	May 2022 – June 2026
SR College — GPA: 8.7/10	Warangal, TG
<i>Maths, Physics, Chemistry</i>	May 2020 – April 2022

SKILLS

Programming & Querying: Python • SQL • Java

Data Science & Machine Learning: scikit-learn • XGBoost • Classification • Regression • Feature Engineering • Model Evaluation • Cross-Validation

Deep Learning & Computer Vision: PyTorch • CNNs • U-Net • Attention Mechanisms • Medical Image Segmentation

Data Analysis & Visualization: Pandas • NumPy • Matplotlib • Seaborn • Plotly

NLP: NLTK • Text Feature Engineering • Lexical Analysis • Semantic Search

LLMs & Applied AI: Prompt Engineering • RAG • LangChain

Databases: MySQL • SQLite • Vector Databases (Qdrant)

Tools: Git • Linux • Jupyter Notebook • VS Code

Soft Skills: Problem Solving • Communication • Analytical Thinking • Fast Learner • Collaboration • Adaptability • Attention to Detail

PROJECTS

- Confidence & Clarity Detection in Interview Responses** | [Python, scikit-learn, Pandas, NLTK, Feature Engineering] | GitHub
- Engineered 20+ linguistic features (sentence coherence, filler word frequency, lexical density) from 1,200+ transcribed responses, identifying top 8 predictors via EDA and correlation analysis.
 - Trained and evaluated 5 classification models (Logistic Regression, Random Forest, XGBoost, SVM, KNN) achieving 91% accuracy using stratified cross-validation.
- Dental Radiograph Segmentation using Hybrid Deep Learning** | [Python, PyTorch, Computer Vision, Data Analysis] | GitHub
- Developed a medical imaging pipeline to preprocess and analyze 3,000+ dental radiographs, applying 8+ augmentation techniques to improve model generalization by ~15%.
 - Optimized segmentation performance achieving 87% Dice Score and 82% IoU across complex dental structures through systematic experiment tracking.
- Brain Tumor MRI Segmentation & Classification** | [Python, Deep Learning, Medical Imaging, Data Science]
- Built an end-to-end data pipeline for MRI analysis across 6,000+ images spanning 4 tumor classes, performing 20+ hyperparameter tuning experiments.
 - Achieved ~90% Dice score and ~95% classification accuracy, reducing false-positive rate by ~12% through ensemble-based post-processing.

CERTIFICATIONS

- Data Analyst 101 — Simplilearn (2025)
- Python Mega Course — Udemy (2023)
- Data Visualization — Kaggle (2026)

PUBLICATIONS

- ResDenseAtt-UNet: An Attention-Enhanced DenseUNet Model for Dental Segmentation* — ICSSCC (2024)
- Real-Time Web Chat Application with Live Performance Metrics* — IEEE (2024)
- Healthcare AI - Dental Assessment Chatbot* — IEEE (2025)